

Coefficient Activity and Data Selection in Verifiable-Reward RL

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Code and logs are included in the anonymized supplementary archive.

Abstract

RL with verifiable rewards (RLVR) often treats prompt curricula and failure recycling as estimator-agnostic data interventions. We ask a narrower question: what task-level activity does a particular group estimator make available to those interventions? For the zero-stabilizer form of practical MaxRL with N i.i.d. binary rollouts, the expected absolute coefficient mass is exactly $2[1 - (1 - p)^N - p] = 2(\text{pass}@N - \text{pass}@1)$. This yields a rollout-aware activity score, recovers $p(1 - p)$ at $N = 2$, and, in this idealization, reveals that dropping all-fail groups targets truncation order $N - 1$ rather than N . In a fresh 20-seed Acrobot tournament with a common nominal transition budget at deployed $N = 16$, u_{16} improves target-uniform AUC over the common-scaffold $p(1 - p) = u_2$ score by $+.0480$ (paired bootstrap 95% CI $[+.0209, +.0738]$, exact sign-flip $p = .0034$) and over uniform by $+.0419$ (Holm-adjusted $p = .0016$). A separate 80-pair paid-probe comparison did not support u_{16} over source-faithful ProCuRL environment selection ($\Delta = .0049$, $p = .0515$; SESOI .02): probes consumed about 93.2% of paid transitions, and every probed arm trailed ordinary uniform by about .31 AUC. A fresh 24-block exact-probability Digits counter-test did not support the internally frozen sampler-by-estimator interaction ($p = .350$): RLOO reversed its predicted ordering, and both matched samplers underperformed uniform. In a balanced MaxRL-versus-GRPO maze factorial, an external execution record reports that a time-integrated contrast was fixed before a six-block fresh wave. MaxRL retained more pass@8 AUC in 6/6 blocks under each sampler (sign $p = .031$ per sampler), although the locking object and checkpoint trajectories are absent from this release. Finally, the reported three-seed Countdown recycling-package aggregate moves mean@16 from .278 to .324 while the logged VERL bootstrap best@16 coverage proxy moves from .541 to .492; this proxy is not standard unbiased pass@16. Together, the positive and negative tests show that exact coefficient activity can generate useful curriculum hypotheses, but is not a universal selection objective.

1 Introduction

RLVR converts automatically checkable outcomes into policy updates, but every update first pays for a group of model rollouts. Two data-level interventions try to make that budget productive. A *difficulty curriculum* samples tasks believed to be learnable now; *failure recycling* relabels a failed trajectory as a verified success for a task it actually achieved. Both are often evaluated as portable modules, separately from the estimator that turns their sampled data into gradients.

That separation is fragile. Group-relative estimators emit no contrast when all N samples agree, while success-conditioned estimators may drop all-fail groups entirely. Variance normalization can continue weighting near-mastered prompts after a success-conditioned estimator has released them. Relabeling changes the achieved-task distribution toward outputs the current policy already reaches. A sampler, recycler, and estimator can therefore move training mass toward different parts of the task pool even when all are described as improving efficiency.

We organize these interactions around the expected absolute coefficient mass emitted by the deployed estimator before accounting for score-gradient direction or magnitude. For practical MaxRL [1], this mass is

$$A_N(p) = 2\{1 - (1 - p)^N - p\} = 2(\text{pass}@N - \text{pass}@1).$$

Its two zeros separate a dead zone, a learnable region, and a mastered tail. Its peak moves toward harder prompts as the rollout count grows. Coefficient mass is not a proof of learning progress, but it is estimator-side activity a task sampler can predict before purchasing a group of rollouts.

This view yields three contributions:

1. **Exact rollout-aware coefficient activity.** We derive A_N , its peak and asymptotics, and show that the practical drop-all-fail estimator targets truncation order $T = N - 1$, not N . A fixed-completion sweep checks its nontrivial N dependence. The identity is an estimator-side diagnostic, not a learning theorem.
2. **Controlled curriculum tests with mixed outcomes.** In fixed-pool Acrobot, u_{16} beats both $p(1-p) = u_2$ and uniform sampling. In a separate paid-probe Acrobot family, registered u_{16} superiority over ProCuRL selection is unsupported and probe cost dominates all three probed arms. In an exact-probability Digits counter-test, the internally frozen interaction fails, RLOO reverses the predicted ordering, and neither matched sampler beats uniform. This positive-negative pair defines the scope of the activity score.
3. **Qualified neural diagnostics.** At common optimizer settings, the externally recorded fresh maze wave has higher MaxRL than GRPO coverage in all six blocks under each sampler. A reported three-seed Countdown aggregate shows higher average correctness but lower VERL bootstrap best@16 under a deployed recycling package. We preserve the registration, artifact, metric, and dose limitations of both results.

The paper is deliberately narrower than the full project record. The main text contains the exact identity and a compact evidence chain. Exploratory mechanisms, failed hypotheses, robotics pilots, gate studies, and implementation detail remain in the appendix and artifact rather than becoming additional contributions.

2 Coefficient activity under practical MaxRL

Setup. For task x , let $r_i \in \{0, 1\}$ be N conditionally independent rollout rewards with pass rate $p = p_\theta(x)$ and $K = \sum_i r_i$. Let $s_i = \nabla_\theta \log \pi_\theta(y_i | x)$ be the response score and $\hat{g} = \sum_i w_i s_i$ the group update. Then $\text{pass}@N = 1 - (1-p)^N$. We analyze the zero-stabilizer idealization of the released practical centered MaxRL convention, which assigns

$$w_i = \frac{r_i}{K} - \frac{1}{N} \quad \text{when } K > 0, \quad w_i = 0 \quad \text{when } K = 0.$$

All-pass groups also have zero contrast. The released implementation uses a finite denominator stabilizer ε ; for $K > 0$ this multiplies the ideal coefficients by $K/(K + N\varepsilon)$. Thus the identities below are exact for the stated zero- ε convention rather than for its finite- ε magnitude.

Lemma 1 (Practical truncation order). *The expectation of the drop- $K = 0$ practical estimator equals the gradient of the $(N - 1)$ -term truncated maximum-likelihood objective, not the N -term objective.*

For $J_T(p) = -\sum_{t=1}^T (1-p)^t/t$, direct conditioning gives

$$\mathbb{E}\hat{g} = \left(\frac{1 - (1-p)^N}{p} - (1-p)^{N-1} \right) \nabla p = \left(\sum_{j=0}^{N-2} (1-p)^j \right) \nabla p = \nabla J_{N-1}.$$

The correction matters whenever a theoretical truncation parameter is mapped to a deployed group size. Details and the non-dropped control-variate convention are in App. A.

Proposition 1 (Exact coefficient mass). *The expected absolute coefficient mass of the practical estimator is*

$$A_N(p) = \mathbb{E} \left[\sum_{i=1}^N |w_i| \right] = 2\{1 - p - (1-p)^N\} = 2(\text{pass}@N - \text{pass}@1).$$

For a mixed group with $0 < K < N$, successes and failures each contribute $(N - K)/N$ absolute mass. Therefore

$$A_N(p) = \frac{2}{N} \mathbb{E}[(N - K)\mathbf{1}\{K > 0\}] = 2\{(1-p) - (1-p)^N\}.$$

This identity concerns coefficients, not gradient norm, signal-to-noise ratio, or expected improvement: score-gradient geometry can change its downstream effect.

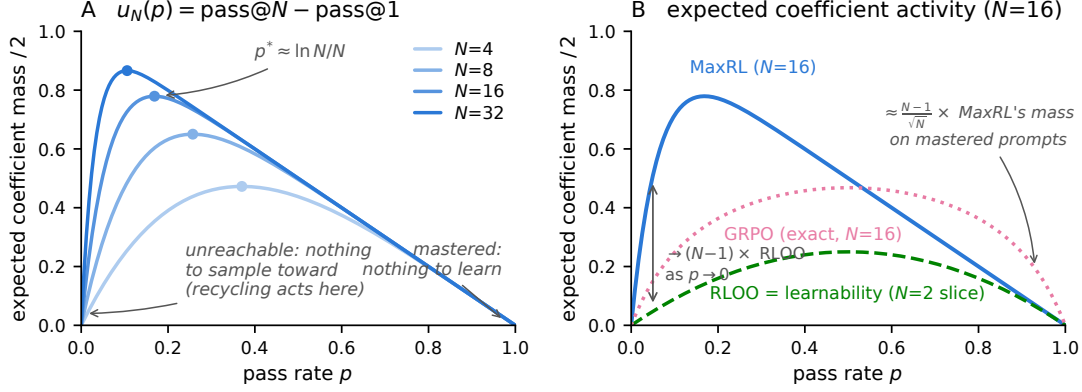


Figure 1: **Estimator-conditioned activity.** *Left:* $u_N = A_N/2 = \text{pass}@N - \text{pass}@1$ for several rollout counts; its peak is derived rather than tuned and moves toward harder tasks as N grows. *Right:* exact expected half-mass at $N = 16$ under zero-stabilizer MaxRL, standard gradient-averaged RLOO, and the zero-stabilizer sample-SD-normalized GRPO convention. Coefficient mass is sign-blind, so the curves motivate rather than prove the empirical coverage ordering.

Table 1: **Post-guidance CPU check.** Mean normalized pass-rate AUC differences at fixed sampled-completion budget. Seed signs are descriptive; the sweep was not pre-registered.

N	2	4	8	16	32
$u_N - p(1 - p)$	.0000	+.0307	+.0920	+.1526	+.1909
$u_N - \text{uniform}$	-.0106	-.0032	+.0309	+.0453	+.0836
wins vs. $p(1 - p)$	identical	8/8	8/8	8/8	8/8

The bridge to the expected update is nevertheless exact. With $\mu_+ = \mathbb{E}[s_i \mid r_i = 1, x]$ and $\mu_- = \mathbb{E}[s_i \mid r_i = 0, x]$,

$$\mathbb{E}[\hat{g} \mid x] = u_N(p)(\mu_+ - \mu_-), \quad u_N(p) = A_N(p)/2.$$

Thus u_N is the estimator-controlled scalar factor in the expected update; the conditional score geometry remains outside the mass calculation.

Consequences. The derivative of $u_N(p) = 1 - p - (1 - p)^N$ gives the unique peak

$$p_N^* = 1 - N^{-1/(N-1)} \simeq \frac{\log N}{N}.$$

Thus the relevant intermediate difficulty depends on rollout budget. At $N = 2$, $u_2 = p(1 - p)$, coinciding with a common learnability score. As $p \rightarrow 0$, MaxRL carries $(N - 1)$ times the coefficient mass of standard $1/N$ -averaged RLOO. As $p \rightarrow 1$, MaxRL and RLOO both decay at first order in $1 - p$, whereas sample-SD-normalized GRPO allocates relatively more mass to near-mastered prompts (App. A).

Fixed-completion check. We tested the arbitrary- N prediction on a 36-task CPU skill-chain testbed with $N \in \{2, 4, 8, 16, 32\}$, eight paired seeds, no recycling, and exactly 51,200 sampled completions per cell. Teacher decay was normalized per completion so changing N did not silently change its memory horizon.

The $N = 2$ identity reproduces bit-for-bit. For every $N > 2$, the rollout-aware utility beats the score-only $p(1 - p)$ comparator in all paired seeds. The mean gap increases across tested N , but only 5/8 seed trajectories are monotone, so we do not claim a scaling law. Nor does u_N universally beat uniform: the mean contrast becomes positive only at $N \geq 8$. The paired positive-seed counts are 3/8, 3/8, 8/8, 7/8, and 7/8 as N increases. This is evidence for rollout awareness in one co-designed testbed, not a blanket curriculum advantage or an algorithm-level comparison with prior work.

Exact-probability counter-test. The stronger test does not generalize the sampler mapping. Under an internally hashed local protocol without an immutable public pre-execution commit, a fresh 24-block Digits contextual bandit computes the policy’s correct-class probability exactly before every update. Its internally frozen estimator-by-sampler interaction is $+0.01589$ (95% CI $[-0.01686, +0.04712]$, exact sign-flip $p = .350$; 15/24 positive) and is not supported. MaxRL favors u_8 over $p(1 - p)$ by $+0.20842$, but RLOO also favors u_8 , reversing its prediction; both estimator-matched samplers are below uniform. The activity algebra therefore generates testable hypotheses, not a universal learnability or curriculum-optimality law.

3 From mass to interventions

Task sampling. FrontierMax operationalizes the identity without changing the within-group estimator. Each task maintains discounted Beta evidence, draws a posterior pass rate $\tilde{p}_x$, and samples approximately as

$$q(x) = \frac{\epsilon}{M} + (1 - \epsilon) \frac{[u_N(\tilde{p}_x)]^\gamma}{\sum_{j=1}^M [u_N(\tilde{p}_j)]^\gamma}.$$

The floor ϵ , concentration γ , decay, and posterior estimator remain empirical choices. The theory contribution is the utility and its N -dependence, not a parameter-free scheduler.

Failure recycling. For an all-fail request, a verified achieved goal can create an auxiliary update where repeating that request would emit none. This requires an exact verifier and rewriting any goal-bearing conditioning. Even then, the achieved goal is selected from a failed group and generally does not yield an unbiased fresh-task policy gradient. We treat recycling as a verified auxiliary update. It changes both update dose and direction, while the achieved-goal distribution is biased toward what the policy already reaches [24, 25].

Predictions. The mass curves motivate, but do not prove, two empirical expectations. First, MaxRL and variance-normalized GRPO can exhibit different coverage orderings under a common sampler; this need not imply a sampler-by-estimator interaction. Second, the deployed recycling package may trade a coverage proxy for mean accuracy when it concentrates updates on reachable outputs. Both require controlled measurement.

4 Evidence

4.1 Externally recorded fresh-wave maze result

Design. The testbed is a 1.26M-parameter transformer trained on procedurally generated 17×17 mazes. Each balanced block shares a seed, warm start, task pool, and evaluation schedule across $\{\text{MaxRL}, \text{GRPO}\} \times \{\text{uniform}, \text{FrontierMax}\}$. The outcome is time-integrated pass@8 coverage relative to the warm-start checkpoint, not final reward. Both estimators use identical deployed optimizer settings; we did not separately tune learning rate per estimator.

Each run uses $N = 32$, eight prompt groups per training step, and 250 steps. Evaluation occurs every 25 steps on 16 held-out mazes per level. We define the coverage AUC as the mean pass@8 over steps 25 through 250 minus the warm-start pass@8 value.

The first factorial rejected its stronger externally recorded endpoint claim: only 3/6 uniform blocks and 4/6 FrontierMax blocks had the predicted sign. We retracted the earlier zero-exception cohort claim. Time-integrated coverage was positive in all six blocks under each sampler in that wave, but this outcome had been discovered exploratorily. The external execution record says it was specified as the primary of a second wave before running new seeds, including its falsification rule. The corresponding locking commit object is not vendored in this checkout.

Result. The wave-2 primary is positive in all six fresh blocks under uniform sampling and all six under FrontierMax (exact two-sided sign $p = .03125$ per sampler). Averaging the correlated sampler observations within each independent block yields mean $+0.0195$ with a post-hoc 95% t interval $[+0.0115, +0.0275]$; every

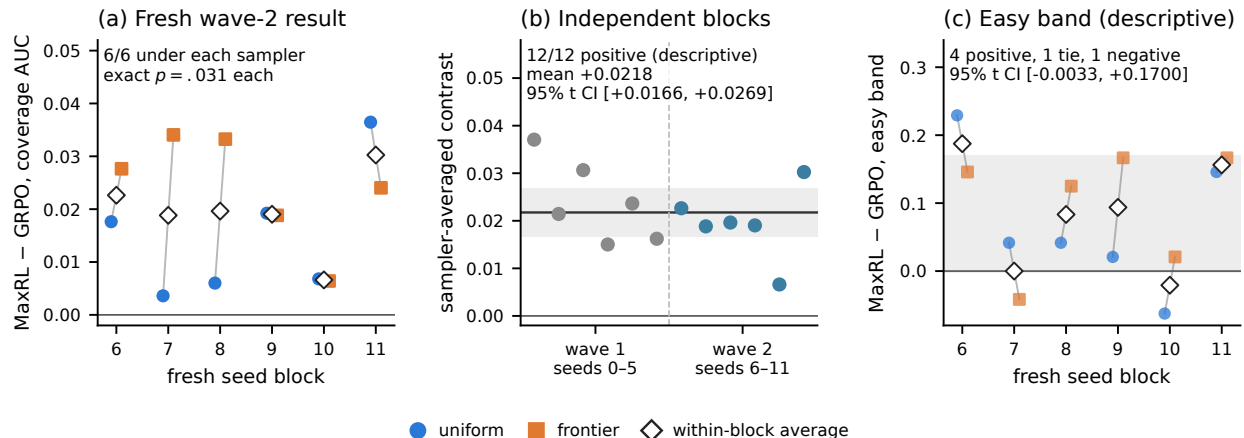


Figure 2: **Maze factorial at the independent seed-block unit.** (a) The externally recorded fresh-wave result is positive in 6/6 blocks under each sampler (exact sign $p = .03125$ each). Connected observations share a block. (b) Sampler-averaging gives wave-2 mean $+0.0195$ and post-hoc 95% CI $[+0.0115, +0.0275]$; 12/12 cross-wave block averages are descriptive. (c) Easy-band localization is suggestive: four positive blocks, one tie, one negative, and an interval containing zero.

leave-one-block-out interval remains above zero. Across both waves, all 12 block averages are positive, but that statement is descriptive because wave 1 selected the outcome.

The data support an estimator-conditioned coverage ordering at common deployed optimizer settings, not universal estimator superiority or the strong mechanism account we initially expected. The recorded pair-level easy-band criterion was met, yet block aggregation leaves four positive, one tied, and one negative easy-band contrast with an interval crossing zero. Localization therefore remains suggestive. We froze an AUC robustness multiverse, but the checkpoint trajectories required to evaluate its warmup, integration, and horizon variants remain external (App. C).

4.2 Fresh fixed-pool Acrobot score test

We next isolate the arbitrary- N score on official Gymnasium **Acrobot-v1**. Eight nested tip-height predicates share one task-blind H64 actor (640 parameters); the hardest predicate is exactly native Acrobot success. All arms use practical MaxRL with $N = 16$, no hindsight, learning rate 3×10^{-4} , and a nominal two million paid environment transitions. Twenty fresh paired seeds compare uniform sampling, the common-scaffold $p(1-p) = u_2$ score, and u_{16} under identical discounted Beta tracking. The internally frozen primary is target-uniform normalized transition-AUC for $u_{16} - p(1-p)$, with support requiring a positive exact two-sided sign-flip test at .05 and a point estimate of at least $+0.01$.

Table 2: **Fresh Acrobot tournament.** Paired differences use 20 fresh seeds under the source/runtime-locked protocol. The two uniform comparisons form one Holm family.

contrast	mean Δ	95% boot. CI	exact p	Holm p	decision
$u_{16} - p(1-p)$	+.0480	[+.0209,+.0738]	.0034	—	supported by rule
$p(1-p) - \text{uniform}$	-.0062	[-.0226,+.0122]	.5078	.5078	not supported
$u_{16} - \text{uniform}$	+.0419	[+.0218,+.0606]	.0008	.0016	supported

The arm means are .64523 (uniform), .63907 ($p(1-p)$), and .68711 (u_{16}). The primary clears both frozen criteria; 15/20 paired differences are positive. The result therefore supports the deployed- N score shape against its $N = 2$ ablation in this fixed family, while the $p(1-p)$ arm itself is not supported over uniform by the frozen secondary test.

The mean $u_{16} - p(1-p)$ contrasts remain descriptively positive on sampled-group ($+0.05294$) and optimizer-

update (+.05366) axes. u_{16} also receives 2.40 fewer groups and 3.42 fewer updates per million transitions, so it did not win by receiving more updates; these non-confirmatory axes do not identify a causal mechanism. Native-success transition-AUC is .31085, .30521, and .37615 across the three arms. Observed mass means are close between adaptive arms (.7344 versus .7490 per group; 115.57 versus 115.90 per million transitions), and both descriptive paired intervals include zero; no equivalence test was frozen. App. C records the audited but RNG-confounded precursor separately.

4.3 Paid-probe ProCuRL confirmatory comparison

A separate frozen family tests source-faithful ProCuRL environment-selection semantics, not merely the $p(1 - p)$ score on a shared posterior. On the same eight-task Acrobot family and 640-parameter practical-MaxRL learner, 80 paired seeds compare ProCuRL selection, range-matched u_{16} , a paid-probe uniform sham, and ordinary uniform. The three probed arms refresh 20-episode-per-task estimates every 5,120 student transitions and charge every probe transition to the common two-million paid budget. The internally frozen fixed-paid-AUC contrast u_{16} -ProCuRL is +.00489 (paired $t_{79} = 1.977$, $p = .0515$; percentile-bootstrap 95% CI [+ .00011, + .00973]). Because $p > .05$ and the point estimate is below the .02 SESOI, the frozen rule is unsupported. Neither adaptive-versus-sham secondary rejects after Holm correction (adjusted $p = .4568$ and .4547); this non-rejection is not an equivalence claim. Probes consume about 93.2% of paid transitions; the three probed arms trail ordinary uniform by $-.314$, $-.309$, and $-.312$ AUC, respectively (all Holm-rejected). Thus this cadence diagnoses probe-cost domination, not general ProCuRL inferiority or a reproduction of its complete PPO system. This 80-pair family and multiplicity plan are separate from Table 2; full accounting is in App. C.

4.4 Recycling-package sharpening

Design. We fine-tune SmolLM2-360M on exact-verifier Countdown tasks with $N = 16$. The retained comparison is a three-seed aggregate on a fixed 128-task tier-1 evaluation set at step 60. Mean@16 measures average correctness over samples; the logged VERL best@16 metric applies 1,000 with-replacement resamples of size 16 to each prompt’s 16 binary outcomes and averages whether a resample contains a success. Conditional on c observed successes, it approximates $1 - (1 - c/16)^{16}$. It is a bootstrap coverage proxy, not standard unbiased pass@16 or the indicator that the observed set contains a success [23]. A summary-backed identity audit reports disjoint RL train/evaluation pools, zero SFT overlap for tier 1, and 27/128 exposed tier-0 tasks. Tier 0 is excluded from absolute claims. The source manifests needed to recompute those statements are not in this checkout.

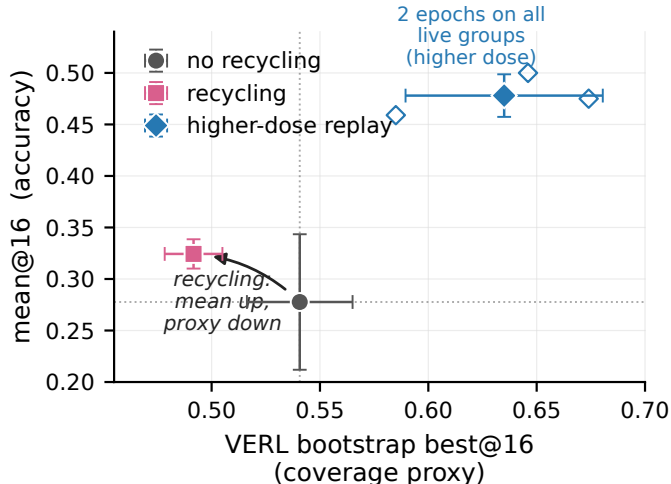


Figure 3: **Countdown accuracy–bootstrap-coverage-proxy tradeoff.** Error bars are sample standard deviations across three seeds; complete three-seed B1/B2 records are not vendored (seed 3 is aggregate-only). The reported recycling-package aggregate lies at higher mean@16 but lower VERL bootstrap best@16. The live-group replay seeds use two optimizer epochs on every live group, a substantially larger dose. This higher-dose alternative improves both metrics but cannot separate dose from update direction.

Result. The reported recycling-package aggregate moves tier-1 mean@16 from $.278 \pm .066$ to $.324 \pm .014$ while bootstrap best@16 moves from $.541 \pm .024$ to $.492 \pm .013$ (mean $\pm$ sample SD across three seeds). The checkout lacks the complete three-seed B1/B2 records, so we do not claim a paired seed test or per-seed sign result. We call the aggregate pattern *recycling-package sharpening*: the reported aggregate is more accurate on average while the logged coverage proxy is lower.

An externally specified control applies two optimizer epochs to every live group with hindsight disabled. It reaches mean@16 $.478 \pm .021$ and bootstrap best@16 $.635 \pm .046$, with all three replay endpoints vendored. Recycling permits at most 12 auxiliary groups per 64 requested groups (18.75%; realized dose can be lower), whereas replay doubles optimizer epochs on every live group. It provides a higher-dose alternative that improves both metrics, without decomposing relabel direction from update dose or justifying a dose-monotonic causal bound.

Two negative results constrain the story. A favorable earlier gate setting used faulty decay code; a corrected, externally specified strong-gate sweep behaved like recycling-off and refuted the claimed monotone gate dial. A separate Jugs suite collapsed under every arm because its learnable region was effectively one template stratum. Neither the gate nor cross-domain generality is a contribution here.

5 Related work

Curricula and rollout allocation. Intermediate-difficulty sampling predates this work. Under ProCuRL’s experimental assumption that the unknown optimal-policy success is one, its proximal score reduces to $p(1-p)$ [7], while SFL [21] and LILO [6] prioritize binary reward variance. Our distinction is convention-specific: $u_N = A_N/2$ is the arbitrary- N expected half-mass of the deployed centered MaxRL rule; centering only when $K > 0$ and zeroing the whole $K = 0$ group induces truncation order $T = N - 1$. Our paid-probe comparison instantiates the published environment-selection semantics on a common learner and shows that, at the frozen refresh cadence, probe accounting overwhelms differences among the adaptive scores. VCRL [11] thresholds realized group variance with memory; classical PLR [12] and ALP-GMM [15] use TD error, staleness, or temporal learning progress. Those scores can react to forgetting; coefficient mass is an instantaneous estimator-activity proxy. PAIRED [13] instead trains an environment adversary and antagonist to generate solvable high-regret tasks, while ACCEL [14] edits previously useful levels. Neither

has a native fixed-pool analogue in our Acrobot threshold family, so we do not present a threshold sampler as a reproduction of either method.

GRESO [8] and DPS [9] select prompts before rollout using historical or predicted reward dynamics. VIP [10] instead varies per-prompt rollout counts to reduce predicted gradient variance. These are especially close operational baselines, but none uses the fixed- N half-mass of the deployed practical MaxRL convention.

HORA [16] reallocates a rollout budget across prompts by varying per-prompt group size to maximize posterior hit utility; VIGOR [17] progressively assigns extra rollouts to high-variance prompts. DARS [18] and Reinforce-Ada [19] use multi-stage or successive allocation. FrontierMax instead selects the task distribution before generation while holding N fixed. BBG [3] gates updates by posterior marginal pass@ k contribution, and boundary-aware Curriculum RL [5] adds external guidance near or beyond an estimated pass@ k reasoning boundary. These criteria target hit probability, reward or gradient variance, adaptive exploration, boundary metrics, or signal creation—not practical-MaxRL coefficient mass. RL2ML [20] further separates finite-rollout objectives, update scale, variance, and metric gain. Its $T = N$ alignment uses the direct estimator or a score-baseline control variate retained on $K = 0$; our $T = N - 1$ statement is for the deployed hybrid convention that centers only for $K > 0$ and zeros $K = 0$.

Coverage and hindsight. RLVR can improve pass@1 while reducing high- k coverage [2, 3]. Entropy collapse supplies complementary evidence of reduced exploration [4]; Pass@ k objectives address the metric directly [22]. We study whether an external curriculum changes that coverage under two estimators. Hindsight experience replay [24], goal-conditioned supervised learning [25], and curriculum-guided relabeling establish that achieved-goal selection changes the training distribution. Relabeling also appears with executable verification in program synthesis [26] and learned-judge rewards in VLA post-training [27]. We report a mean-up, bootstrap-coverage-proxy-down aggregate from cross-task relabeling in one RLVR model and task family, not a priority claim or a general theorem about hindsight.

6 Limitations

Coefficient mass ignores score-gradient direction, parameter sharing, optimizer state, task transfer, and non-i.i.d. rollouts. It is a sampler-facing surrogate, not a guarantee of loss reduction or metric improvement. The fixed-completion N -sweep is post-guidance and uses one co-designed synthetic chain. The exact-probability Digits counter-test is a contextual bandit rather than trajectory RL, and its internally hashed local lock lacks an immutable public pre-execution commit establishing timing. The fresh V2 Acrobot score test uses one fixed eight-threshold family, one 640-parameter learner, and no held-out task distribution or native named-method implementation. Its 20-seed choice was inherited from a historical precursor rather than set by prospective power analysis; its exact paired test also requires sign exchangeability under the sharp null. The separate 80-pair comparison implements ProCuRL’s environment-selection semantics on that common learner, not its full PPO system. Its frozen 5,120-transition refresh cadence spends about 93.2% of paid transitions on probes, so the result cannot establish general ProCuRL inferiority or performance under a cheaper probe design. Complete-group execution produces bounded transition overshoot, and retained evaluation records do not include per-episode trajectories. The maze result uses a small transformer, one task family, and common rather than estimator-tuned learning rates; its easy-band mechanism remains descriptive. The Countdown result has three seeds and one 360M model, and its single-seed per- k curve is exploratory. A controlled GSM8K cell missed its externally recorded treatment-delivery gate by .00148 and is inconclusive by design.

The external execution record describes the maze wave-2 outcome and Countdown controls as specified before execution, but their locking commit objects are not vendored in this checkout. The release can audit the stored protocols and verdicts, not independently verify their timing. The two Acrobot locks and digests are vendored and internally bind source, runtime, gates, and artifacts, but this checkout likewise lacks an immutable public pre-execution commit establishing either study’s timing.

The Countdown coverage headline is limited further by metric provenance: the available scalar is VERL bootstrap best@16, not standard unbiased pass@16. Missing per-task outcomes prevent recomputation and leave the size of the metric mismatch unknown.

Artifact coverage is incomplete. The generated registry contains 562 run records supported by local artifacts, receipts, or summaries: 441 are Acrobot. Thirty-three rows point to individual vendored run-

level files, 121 Acrobot rows point into vendored aggregates retaining per-run diagnostics, 320 paid-probe confirmation rows are bound by per-run receipts to an external raw artifact, and 88 are explicitly summary- or external-backed. Complete B1/B2 Countdown records, including seed 3, and per-task outcomes are missing. A clean 101-task tier-0 analysis cannot be reconstructed without those outcomes or compatible checkpoints. The compact Digits release includes all block contrasts and a content manifest for 2,904 scientific files, but its 5.08 GB full ledger/checkpoint payload remains local without a download URI; historical full replay therefore requires regeneration or access to that payload. These gaps limit auditing and preclude a broad neural claim.

7 Conclusion

The estimator belongs in the evaluation of data-selection design. Its coefficient algebra characterizes which tasks can emit activity at a chosen rollout count, and its weighting can change what coverage a sampler removes or preserves. The exact MaxRL mass identity supplies a rollout-aware activity proxy; the fresh Acrobot tournament shows that its deployed- N score can outperform both the $N = 2$ score and uniform sampling in a fixed classical-control family. The separate paid-probe comparison does not support u_{16} over ProCuRL selection and exposes probe-cost domination at its frozen cadence, while the fresh Digits counter-test refutes a universal estimator-to-sampler mapping. The externally recorded maze wave shows a coverage ordering at common optimizer settings, while the reported Countdown aggregate has higher mean accuracy but a lower logged bootstrap coverage proxy under a deployed verified-recycling package. Curricula and recyclers should therefore be evaluated with the estimator they ship on, with $\text{pass}@k$ computed from retained task outcomes and reported beside mean accuracy, and coefficient mass treated as a testable allocation hypothesis rather than a universal learning law.

Reproducibility statement

The repository includes exact-enumeration tests, a one-command artifact check, structured seed-block analyses, stored protocol and verdict documents, figure derivation scripts, a source-locked 60-run Acrobot tournament with deterministic retained-ledger reanalysis, a separately locked 320-run paid-probe comparison with an external-raw receipt manifest, and a generated 562-row registry. Registration timing objects that remain external are disclosed rather than treated as locally auditable. The independent unit is stated for every central result; known raw-data gaps and Countdown SFT overlap are disclosed in App. D.

AI use statement

AI assistants supported planning, implementation, debugging, literature discovery, statistical and artifact auditing, figure preparation, and manuscript editing. The authors selected the questions and decision rules, ran the studies, verified derivations, code, numerical results, and citations, and take responsibility for every claim.

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A Proof details

Truncation order. For $K > 0$, conditioning on one sampled response and using exchangeability gives

$$\mathbb{E}\hat{g} = \left(\frac{1 - (1-p)^N}{p} - (1-p)^{N-1} \right) \nabla p.$$

Since $1 - (1-p)^N = p \sum_{j=0}^{N-1} (1-p)^j$, subtracting the final term leaves $\sum_{j=0}^{N-2} (1-p)^j \nabla p = \nabla J_{N-1}$. If the centered control-variate term is retained on $K = 0$, its contribution is $(1-p)^{N-1} \nabla p$. The expectation becomes $\sum_{j=0}^{N-1} (1-p)^j \nabla p = \nabla J_N$. Thus the direct/non-dropped convention has order N , while the deployed hybrid convention above has order $N-1$; theorem and implementation must use the same convention. At audited MaxRL source commit [7197bbb](#), the released path computes $(r_i - \bar{r})/(\bar{r} + \epsilon)$ and zeros all-failure groups; after the deployed gradient average, its $\epsilon \rightarrow 0$ coefficients match this convention up to common scale. The parsed truncation-order setting is inactive on that path.

Coefficient mass. For $0 < K < N$, each success has coefficient $(N-K)/(KN)$ and each failure has coefficient $-1/N$. The two classes each contribute $(N-K)/N$ in absolute value. For $K \in \{0, N\}$ the deployed centered estimator contributes zero. With $K \sim \text{Bin}(N, p)$,

$$\begin{aligned} A_N(p) &= \frac{2}{N} \sum_{k=1}^{N-1} (N-k) \binom{N}{k} p^k (1-p)^{N-k} \\ &= \frac{2}{N} \{ \mathbb{E}[N-K] - N \Pr[K=0] \} \\ &= 2\{1-p - (1-p)^N\}. \end{aligned}$$

The peak follows from $u'_N(p) = N(1-p)^{N-1} - 1$ and strict concavity for $N > 1$.

Estimator limits. Under the deployed $1/N$ gradient-average normalization, RLOO expected half-mass is $p(1-p) = u_2(p)$, so $u_N(p)/u_2(p) \rightarrow N-1$ as $p \rightarrow 0$. Near mastery, both are first order in $q = 1-p$. With sample-SD normalization, GRPO group mass is proportional to $\mathbb{E}[\sqrt{K(N-K)}]$ up to deployed normalization constants; its relative tail scaling is verified by exact enumeration in `verify_guidance_math.json`. These are coefficient statements, not gradient-direction statements.

Dependent rollouts. The per-group identity does not require independence. For any binary group,

$$\mathbb{E} \sum_i |w_i| = 2\{\Pr(K \geq 1) - \mathbb{E}[K]/N\}.$$

With a common marginal p , the second term is p ; independence is used only to replace $\Pr(K \geq 1)$ by $1 - (1-p)^N$. In a post-guidance exact beta-binomial CPU stress test, count-distribution enumeration matched the general identity to 2.58×10^{-13} . At $\rho = .10$, the activity peak moved from .169 to .234 for $N = 16$ and from .106 to .186 for $N = 32$. This dependence model shows how increasing positive correlation can lower activity inside that family; pairwise correlation alone does not determine a general joint law. The result is a scope diagnostic, not evidence that coefficient mass causes learning or that beta-binomial dependence describes neural rollouts.

B Algorithms and estimator contracts

Posterior teacher. Each task stores discounted successful and failed rollout counts. A Thompson draw $\tilde{p}_x$ is mapped through u_N and mixed with a uniform floor. Teacher state, sampler cursor, and checkpoints restore together. Observations use pre-relabel rewards so recycled outcomes do not inflate the requested task’s posterior.

Relabeling. A parser proposes an achieved goal from a failed response. The environment’s exact verifier must accept the response under that goal. Prompt conditioning and explicit response goal declarations are rewritten; answer spans, arithmetic, decimals, signed values, and unknown contexts are preserved. The reported-run implementation is frozen, while a conservative release wrapper and 26 CPU regression/property tests protect future runs. Relabeled groups remain auxiliary selected-data updates; no importance ratio converts them into fresh on-policy target samples.

C Experimental protocols and secondary results

Fixed-completion N -sweep. The environment contains three nested 12-level chains and ten actions. Every cell runs exactly 51,200 completions, with arms for uniform, $p(1 - p)$, and u_N sampling under the practical MaxRL estimator, no hindsight, and eight paired seeds. Checkpoints share a completion clock. The post-guidance analysis is descriptive.

Digits factorial (internally frozen negative). On a stored 1,077/360/360 train/development/test split of `sklearn` Digits, a 4,810-parameter float64 policy recomputes each example’s exact correct-class probability before every update. We crossed practical MaxRL and RLOO with uniform, $p(1 - p)$, and u_8 sampling. Five-rate development selected learning rate .1 for MaxRL, RLOO, and the common-rate check. Across 24 fresh paired blocks, the internally frozen interaction was +.01589 (paired bootstrap 95% CI $[-.01686, +.04712]$, exact two-sided sign-flip $p = .350$; 15/24 positive), so it was not supported despite passing the treatment-delivery gate. MaxRL did favor u_8 over $p(1 - p)$ by +.20842 (CI $[+.16791, +.24744]$; 23/24), but RLOO’s frozen prediction reversed: $p(1 - p) - u_8 = -.17665$ (CI $[-.22042, -.13593]$; 0/24). Moreover, the matched samplers were below uniform for both MaxRL ($-.11279$) and RLOO ($-.37581$). Because every selected rate equaled .1, the tuned and common schedules have byte-identical ledgers and checkpoints and identical scientific summary content after removing phase/authorization labels; they are identity checks, not independent replications. This exact- p contextual bandit therefore supports neither the planned interaction nor a universal curriculum advantage; it shows why expected coefficient mass must remain an estimator-activity proxy rather than a learning law. Its internally hashed local lock is not backed by an immutable public pre-execution commit that independently establishes timing.

Acrobot V3 (historical). The frozen control uses official `Acrobot-v1`, eight nested thresholds from -1.5 through native success at 1.0 , one shared H64 actor, practical MaxRL, $N = 16$, and no hindsight. Uniform and u_{16} receive 20 paired seeds and a nominal two million actual transitions per run, completing the final group. Evaluation uses 32 shared trajectories total, reused across all eight predicates, at initialization and each 100,000 transitions with fixed common random numbers. The internally frozen primary is target-uniform normalized transition-AUC; group diagnostics and evaluation curves for all 40 runs are vendored. A post-hoc audit found cross-seed overlap between actor-initialization and action-sampling RNG roots, so the retained effect and mechanism audit are descriptive rather than clean confirmation. The mechanism audit is deterministic from those records.

Acrobot fixed-pool tournament V2. The replacement reruns uniform and u_{16} and adds $p(1 - p) = u_2$ on fresh logical seeds 20000–20019. All 60 source/runtime-locked runs completed after an outcome-blind development gate; globally disjoint RNG roots and per-run group-diagnostic and aggregate-checkpoint ledgers were retained. The primary $u_{16} - p(1 - p)$ transition-AUC difference is +.04803 (paired bootstrap 95% CI $[+.02094, +.07385]$, exact sign-flip $p = .003361$), clearing the predeclared +.01 estimate filter; Table 2 gives the Holm-controlled uniform comparisons. The p -value tests zero rather than the SESOI, and pairing does not guarantee sign exchangeability. The $p(1 - p)$ arm isolates only the score on our scaffold, not full ProCuRL or SFL.

Paid-probe ProCuRL selection semantics. This is a separate 80-pair family, not an extension of the V2 table. It keeps the same eight thresholds, H64 actor, practical MaxRL estimator, $N = 16$, and 3×10^{-4} learning rate. Its four arms are ordinary uniform; a uniform sham with paid probes; source-faithful ProCuRL selection $\text{softmax}\{20\hat{p}(1 - \hat{p})\}$; and continuous-range-matched u_{16} selection. Probed arms run 20 episodes per

task initially and at every crossed 5,120-student-transition boundary; all student and probe transitions count toward the nominal two-million paid budget. Across seeds 21000–21079, the internally frozen fixed-paid-AUC contrast u_{16} –ProCuRL is $+.004894$ (paired SD $.022139$, $t_{79} = 1.9773$, $p = .05149$; 20,000-resample CI $[+.000110, +.009727]$; one-million sign-flip $p = .05161$). It misses both the $.02$ SESOI and the $.05$ paired- t criterion. The five Holm-family contrasts are ProCuRL–sham $-.001704$ (adjusted $p = .4568$), u_{16} –sham $+.003190$ ($.4547$), ProCuRL–ordinary $-.313774$ (3.47×10^{-66}), u_{16} –ordinary $-.308880$ (9.48×10^{-67}), and sham–ordinary $-.312070$ (4.06×10^{-68}). Mean fixed-paid AUCs are $.33771$ (ProCuRL), $.33942$ (sham), $.65149$ (ordinary), and $.34261$ (u_{16}).

The probed arms average 28.0–28.1 sweeps and 140.5–141.1k student transitions versus 2.003M for ordinary uniform; 93.19–93.21% of their paid transitions are probes. They average 15.4–19.8 optimizer updates versus 259.1 ordinary-uniform updates, with mean paid-budget overshoot 69,960–72,939 transitions versus 2,984. All 320 confirmation and 12 outcome-blind development runs reconcile exactly, all 11 development gates pass, and the common-random-number invariants replay.

An earlier outcome-blind development wave was invalidated before any gate or arm contrast because 73 entropy-sum replays differed by at most 7.28×10^{-12} . Its seeds were burned and its incident receipt is retained; the reported family uses fresh seeds and a replacement lock.

The external confirmatory raw is 1,374,886,097 bytes. Its SHA-256 boundary is:

```
raw: b1f8756c249effab8c77101c8bca73ddf708a5e143c18fe8742fd5712fdd7c12
lock: b7c7f76f6aaffa1fe65557717bfe545f2ec850495d370cd97fe72a8871fc8d0f
analysis: 2010e30b5b15a212e2d6bdfaacd43d2434e5f468a96be613cf59744b9bc2fb38
```

The source bundle carries a 320-record external-raw manifest: compact mode checks its small-artifact bindings, while full verification requires the raw file and replays locked validation. The local lock is mechanical provenance, not an immutable public pre-execution registration.

Post-guidance rollout-allocation factorial. We also compared fixed $N = 16$, published-HORA-style posterior hit allocation [16], and a mass-aware marginal that counts four already-spent probe rollouts. A 2-sampler by 3-allocator CPU factorial uses 16 paired seeds and 51,200 completions per cell. The mass-aware arm improves pass@8 AUC over HORA-style allocation by $+.00676$ (11/16 positive pairs) and over fixed groups by $+.02517$ (14/16). However, it *reduces* realized coefficient mass per completion by $.000306$ relative to HORA-style allocation in all 16 pairs, falsifying the frozen mechanism prediction. Its group sizes are also extreme (cell-average maximum about 76 versus 57 for HORA and 16 fixed). We therefore treat the result as exploratory motivation for posterior-calibration and allocation-cap ablations, not a fourth contribution.

A frozen follow-up crossed two samplers, the two adaptive scores, four caps, and three information sources over 16 paired seeds (50 cells; 800 runs). Restricting to the 32 deployable-information adaptive cells, all 32 adaptive-minus-fixed mean pass@8-AUC contrasts were positive, with 22 unadjusted descriptive Student- t intervals above zero; no multiplicity-adjusted inference is assigned. Yet realized coefficient mass was lower in 28/32 cells, with 24 intervals below zero and none above zero. The remaining 16 oracle-preupdate cells are diagnostic. Thus the expanded result reinforces the learning-mass mismatch rather than a mediation claim. The frozen engineering rule chose cap 32: averaged over samplers for the fresh-group/same-step cells, it reduced maximum group size by 58.07% with an AUC change of $-.00271$ versus uncapped. This is a point-estimate filter for a future pilot, not an equivalence test. The source/runtime lock is internally hashed but lacks a public pre-execution commit establishing timing.

Maze factorials. Wave 1 has six shared seed/warm-start blocks and six arms per block: MaxRL and GRPO under uniform and FrontierMax sampling plus two diagnostic GRPO arms. Its recorded endpoint hypothesis failed. Wave 2 has six fresh blocks and the four core arms. The external execution record names time-integrated pass@8 relative to warm start primary, with at least 5/6 positive blocks required under each sampler. The block analyzer reports sampler-specific exact sign tests, sampler-averaged contrasts, post-hoc intervals, and leave-one-block-out checks. No pooled p -value treats sampler observations as independent.

AUC robustness status. The frozen multiverse covers integration rule, warmup inclusion, horizons, and leave-one-checkpoint-out variants, but all 24 wave-2 checkpoint JSONLs and warm starts remain external. Structured summaries recover the anchors (uniform 6/6, mean $+.01496$; FrontierMax 6/6, $+.02404$; sampler-averaged 6/6, $+.01950$), but cannot identify those variants; the analyzer therefore exits explicitly blocked.

Countdown. The v2 pool uses operand permutation, parenthesization, and exact division. Tier 1 has 128 evaluation tasks and 16 verifier-scored samples per task. B1 disables recycling; B2 admits exact-verifier achieved goals; the committed scoreboard contains three-seed aggregates. ARM B disables recycling and uses two optimizer epochs on every live group; its three individual endpoints are vendored. Headline endpoints are step-60 mean@16 and VERL bootstrap best@16. The latter uses 1,000 with-replacement resamples per prompt and is not standard unbiased pass@16; missing per-task outcomes prevent recomputation. A stored task-identity summary reports zero tier-1 and tier-2 SFT overlap and 27 exposed tier-0 tasks; the source manifests needed to recompute those counts are absent. Tier 0 is excluded from absolute claims.

Negative branches. The artifact retains the failed maze endpoint, gate, Jugs, and robotics branches. GSM8K missed its delivery threshold (.601480 versus $< .60$) and is inconclusive; the single-seed Countdown per- k curve is exploratory.

D Artifact and data accounting

`bash reproduce.sh --build` tests the identities and integrations, validates the registry and stored endpoints, checks hashes, regenerates figures, and builds both PDFs. The registry has 562 records: 94 maze, 20 Countdown, 7 GSM8K, and 441 Acrobot. Of these, 33 point to individual vendored files, 121 Acrobot rows to aggregates with per-run ledgers, 320 paid-probe confirmation rows to per-run receipts for the external raw, and 88 to summaries or external artifacts. Acrobot comprises 40 historical V3, 9 V2 development, 60 V2 confirmation, 12 paid-probe development, and 320 paid-probe confirmation runs; quick and invalid runs are excluded.

Missing external data are maze checkpoints; Countdown task manifests, outcomes, and B1/B2 seed 3; the 1.375 GB paid-probe confirmatory raw (its content hash and per-run receipt manifest ship); and the Digits 5.08 GB replay payload (block contrasts and a content manifest ship). Compact inputs and checksums are in `paper/results/manifest.json` and fail closed.